Problem Set 7

POLS W4732
Fall 2026

Due December 4 before the TA section

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. *Consider a society composed of three groups: R, M, and P. In all groups, there is a unit 1 of voters (they are equally sized). The utility of each group depends on public spending on two policies: p and s (e.g., police and social security). Each group differs on how much they value policy s. The utility of group $J \in \{R, M, P\}$ is:*

$$U_J(p, s) = p(\alpha - p) + \beta_J s \tag{1}$$

Assume that $\beta_R = 0$, $\beta_P = 1$ and $\beta_M \in (0, 1)$, and $\alpha \in (1, 2)$.
Spending on both policies cannot go over the (exogenous) budget 1: $p + s \leq 1$. Two candidates A and B compete to get elected—which is the only thing they care about—by committing to a platform $(p_i, s_i),\ i \in \{A, B\}$ before the election. After observing candidates' platforms, voters vote for one of the two candidates.

1. *Why is the budget constraint binding?*
2. *Find the ideal spending mix on p and s for each group, denoted by $p^*(\beta)$.*
3. *Show that under perfect information and deterministic voting (each voter votes for the candidate whose platform yields the highest payoff), both candidates propose group M's preferred spending mix.*
4. *Substitute the budget constraint into each voter's utility and write down $V_J(p)$, the indirect utility of a voter from group j of a platform p.*

In what follows, assume that candidate A and B are both policy-motivated. Candidate A shares group-R voters' preferences; candidate B shares group-P voters' preferences. Further assume that turnout in groups M and P is fixed at 50%, while turnout in group R is given by the realization of random variable X, uniformly distributed on $[1/2, 1]$.[1]

[1] Alternatively: the probability that no more than $x\%$ of R voters show up is $\frac{x-1/2}{1-1/2} = 2x - 1$.

5. *Show that convergence is* not *an equilibrium outcome.*

6. *Suppose B proposes $p_B = p^*(\beta^P)$. Find the set of platforms which guarantee victory for A (assume that when indifferent, voters from group M vote for A so you can define a closed interval).*

7. *Now suppose that all voters turn out, but that voters in each group vote for candidate B if and only if*
$$V_J(p^A) \leq V_J(p^B) + \xi$$
where ξ is distributed uniformly in $\left[-\frac{1}{2\psi}, \frac{1}{2\psi}\right]$.
Write down $\pi(p^A, p^B)$, candidate A's winning probability as a function of the platforms.

8. *Write down the objective function of each party, and the associated first-order conditions.*

9. *Using the first order conditions, argue that $P_A(p_B)$, A's best reply to p_B is increasing and strictly larger than p^*_M, and that $P_B(p_A)$, B's best reply to p_A is increasing and strictly smaller than p^*_M .*

10. *Suppose that $\alpha = 2$ and $\beta_M = 1/2$. Show that the distance between the equilibrium platforms $p_A - p_B$ decreases with ψ. [Hint: guess that $p_A + p_B = 3/2 = 2p^*_M$, and use the FONC to obtain a simple expression characterizing $\Delta = p_A - p_B$.]*

Exercise 2 (Cheap talk)**.** *In the baseline Crawford–Sobel model, the state ω is uniform on $[0, 1]$, the Receiver's ideal action is ω, and the Sender's ideal action is $\omega + b$.*

1. *Compute the Sender's ex ante expected payoff under babbling and under a two-report equilibrium. Characterize the values of b for which this equilibrium exists.*

2. *When comparing it to the babbling equilibrium, a person says: "Because the cutoff $\omega^* < 1$, the Sender's ex-ante payoff is higher under the two-report equilibrium, whenever it exists." Determine whether this claim is correct.*

3. *A different person claims: "Because the cutoff $\omega^* < 1$, the Sender's payoff is higher under the two-report equilibrium, whenever it exists, for every realization of the state." Determine whether this claim is correct.*